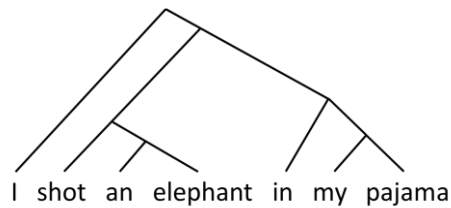
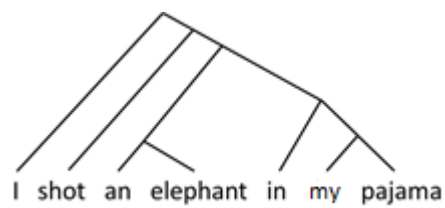


Problem #6: Encoding Trees (20 marks)

Linguistic structures, such as grammatical relations between words in a sentence, are often represented as a *Tree*. For example, the following *parse* or *phrase-structure trees* represent two different interpretations of the same sentence.



Tree-A



Tree-B

The above trees are known as strictly *binary trees* because at every branching point of the tree, there are *exactly two* branches – a left and a right one. Since drawing a tree takes more space and time, trees are often encoded as strings. According to one such encoding scheme for binary trees, the above trees are represented as:

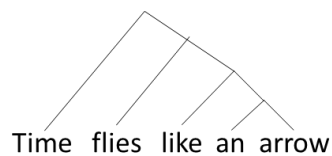
Tree-A: *I 3 shot 1 an 0 elephant 2 in 1 my 0 pajama*

Tree-B: *I 4 shot 3 an 0 elephant 2 in 1 my 0 pajama*

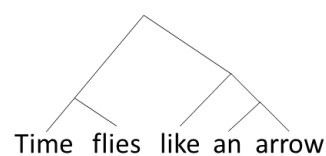
For this encoding scheme, every distinct binary tree has a unique string representation and for every distinct *valid* string representation (like the two examples above), one can also uniquely construct a binary tree. Your task is to figure out this encoding scheme and prove certain facts about it.

Assignment 1:

Shown below are two phrase-structure trees, **Tree-C** and **Tree-D** for two different interpretations of a sentence. Write down the string encodings for **Tree-C** and **Tree-D**. Explain your assumptions.



Tree-C



Tree-D

Assignment 2:

Construct the trees corresponding to the following encoded representations (words have been replaced by letters): **a 0 b 3 c 0 d 1 e 2 f 0 g** **a 0 b 1 c 2 d 3 e 4 f**

Assignment 3:

The following encodings do not represent binary trees. Convert them into valid tree encodings by incrementing or decrementing only one of the numbers by exactly 1.

a 0 b 2 c 0 d 2 e 1 f 0 g **a 1 b 3 c 2 d 1 e 0 f**

There are several constraints that a string must satisfy to be a valid encoding for a binary tree. What constraints do these strings violate, which makes them invalid?

Assignment 4:

Let $a_1 x_1 a_2 x_2 a_3 \dots a_{n-1} x_{n-1} a_n$ be a valid representation of a binary tree for a n word sentence $a_1 a_2 a_3 \dots a_{n-1} a_n$ where $x_1, x_2, \dots, x_{n-1}$ are the numbers separating the words. Prove that

$$2^n - 1 \geq \sum_{i=1}^{n-1} 2^{x_i} \geq \lfloor \log_2 n \rfloor \left\lfloor \frac{n}{2} \right\rfloor$$